Answers

Q1. RGB Channel Isolation

```
img = imread('peppers.png');
redChannel = img; redChannel(:,:,2:3) = 0;
greenChannel = img; greenChannel(:,:,[1 3]) = 0;
blueChannel = img; blueChannel(:,:,1:2) = 0;
subplot(1,3,1), imshow(redChannel), title('Red Channel');
subplot(1,3,2), imshow(greenChannel), title('Green Channel');
subplot(1,3,3), imshow(blueChannel), title('Blue Channel');
```

Q2. Image Arithmetic - Blending

```
img1 = imread('cameraman.tif');
img2 = imread('rice.png');
imgWithTruncate= imadd(img1, img2);
img12 =uint16(img1) + uint16(img2);
mx = max (img12 (:));
mn = min (img12(:));
imgWithNormalization = uint8( 255* double(img12-mn) / double(mx-mn) );
figure, imshow (imgWithTruncate), title ('image With Truncate')
figure, imshow (imgWithNormalization), title ('image With Normalization')
```

Q3. Brightness Control

```
img = imread('cameraman.tif');
brighter = immultiply(img, 1.5);
```

```
darker = imdivide(img, 1.5);

subplot(1,3,1), imshow(img), title('Original');

subplot(1,3,2), imshow(brighter), title('Brighter');
subplot(1,3,3), imshow(darker), title('Darker');
```

Q4. Logical ROI Editing

```
img = imread('liftingbody.png');

mask = im2uint8(roipoly(img));

darkImage = imdivide(img, 1.8);

Msk_Cmp = imcomplement(mask);

darkShip = bitand(darkImage, mask);

bg = bitand(img, Msk_Cmp);

result = bitor(darkShip, bg);

subplot(1,2,1), imshow(img), title('Original');

subplot(1,2,2), imshow(result), title('Edited');
```

Q5. Indexed Image and Grayscale

```
[img, map] = imread('trees.tif');

grayImg = ind2gray(img, map);

figure, imshow(img, map), title('Indexed Image');

figure, imshow(grayImg), title('Grayscale');
```

Q6. Translate Image

```
inputImage = imread('cameraman.tif');

tx = 50; ty = 50;

T = [1 0 tx; 0 1 ty; 0 0 1]';
```

```
tform = affine2d(T);
outputSize = [256, 256];
xLim = [0, outputSize(2) + tx];
yLim= [0, outputSize(1) + ty];
Rout = imref2d(outputSize, xLim, yLim);
outputImage = imwarp(inputImage, tform, 'OutputView' , Rout);
figure, imshow(inputImage), title('Original Image');
figure, imshow(outputImage, Rout), title('Translated Image');
```

Q7. Crop and Flip

```
img = imread('cameraman.tif');
cropped = imcrop(img, [70 30 100 100]);
flipped = fliplr(cropped);
subplot(1,2,1), imshow(cropped), title('Cropped');
subplot(1,2,2), imshow(flipped), title('Flipped');
```

Q8. Scale + Rotate

```
img = imread('cameraman.tif');
S = [0.5 0 0; 0 0.5 0; 0 0 1]';
R = [cosd(45) -sind(45) 0; sind(45) cosd(45) 0; 0 0 1]';
T = S * R;
tform = affine2d(T);
result = imwarp(img, tform);
imshow(result), title('Scaled then Rotated');
```

Q9. Image Complement

```
img = imread('coins.png');
compImg= imcomplement(img);
subplot(1,2,1), imshow(img), title('Original Image');
subplot(1,2,2), imshow(compImg), title('Image Complement');
```

Q10. Indexed Image Conversion

```
[img, map] = imread('trees.tif');
rgb = ind2rgb(img, map);
gray = ind2gray(img, map);
figure, imshow(img, map), title('Indexed');
figure, imshow(rgb), title('RGB');
figure, imshow(gray), title('Grayscale');
```

Q11. Channel Amplification

```
img = imread('peppers.png');
img2 = img;
img2(:,:,1) =  img2(:,:,1) + 50;
figure;
subplot(1,2,1), imshow(img), title('Original');
subplot(1,2,2), imshow(img2), title('Red Channel Amplified');
```

Q12. Flip and Combine

```
img = imread('cameraman.tif');
```

```
flipV = flipud(img);
flipH = fliplr(img);
sumImg = imadd(img, flipH);
figure;
subplot(2,2,1), imshow(img), title('Original');
subplot(2,2,2), imshow(flipV), title('Vertical Flip');
subplot(2,2,3), imshow(flipH), title('Horizontal Flip');
subplot(2,2,4), imshow(sumImg), title('Sum with Horizontal Flip');
```

Q13. Adjust Histogram

```
img = imread('pout.tif');
oldLim = stretchlim(img);
newLim = uint8([50, 200]);
Shrunk_Img = imadjust(img, oldLim, im2double(newLim));
figure
subplot(2,1,1), imshow(img), title('Original Image')
subplot(2,1,2), imhist(img), title('Original Histogram')
figure
subplot(2,1,1), imshow(Shrunk_Img), title('New Image')
subplot(2,1,2), imhist(Shrunk_Img), title('New Histogram')
```

Q14. Adaptive vs Global Histogram Equalization

```
img = imread('coins.png');
histeqImg = histeq(img);
adaptImg = adapthisteq(img);
```

```
figure
subplot(2,1,1), imshow(img), title('Original');
subplot(2,1,2), imhist(img), title('Original');
figure,
imshow(histeqImg), title('Histeq');
subplot(2,1,1), imshow(histeqImg), title('Adapthisteq');
subplot(2,1,2), imhist(histeqImg), title('Original Hist');
figure,
subplot(2,1,1), imshow(adaptImg), title('Histeq Hist');
subplot(2,1,2), imhist(adaptImg), title('Adapthisteq Hist');
```

Q15. Composite Affine Transform

```
img = imread('cameraman.tif');
sx = 1.5; sy = 1.5;
S = [sx 0 0; 0 sy 0; 0 0 1]';
tx = 50; ty = 50;
T = [1 0 tx; 0 1 ty; 0 0 1]';
composite = S * T;
outputsize = [256 * sy, 256* sx];
xLim = [0, outputsize(2) + tx];
yLim= [0, outputsize(1) + ty];
Rout = imref2d(outputsize, xLim, yLim);
tform = affine2d(composite);
result = imwarp(img, tform, 'OutputView', Rout);
figure, imshow(result), title('Composite Transform');
```

Q16. region-based histogram equalization

```
img = imread('liftingbody.png');

mask = im2uint8(roipoly(img));

imgEQ = histeq(img, 256);

Msk_Cmp = imcomplement(mask);

shipEQ = bitand(imgEQ, mask);

bg = bitand(img, Msk_Cmp);

result = bitor(shipEQ, bg);

subplot(1,2,1), imshow(img), title('Original');

subplot(1,2,2), imshow(result), title('Edited');
```

Q17. Chessboard

```
img = zeros(600, 600);

rows1 = [1:100, 201:300, 401:500];

cols1 = [1:100, 201:300, 401:500];

rows2 = [101:200, 301:400, 501:600];

cols2 = [101:200, 301:400, 501:600];

img(rows1, cols1) = 255;

img(rows2, cols2) = 255;

imshow(img)
```

Q18. Square inside Square

```
img = zeros(300, 300);

img(50:250, 50:250) = 255;

img(100:200, 100:200) = 0;

imshow(img)
```